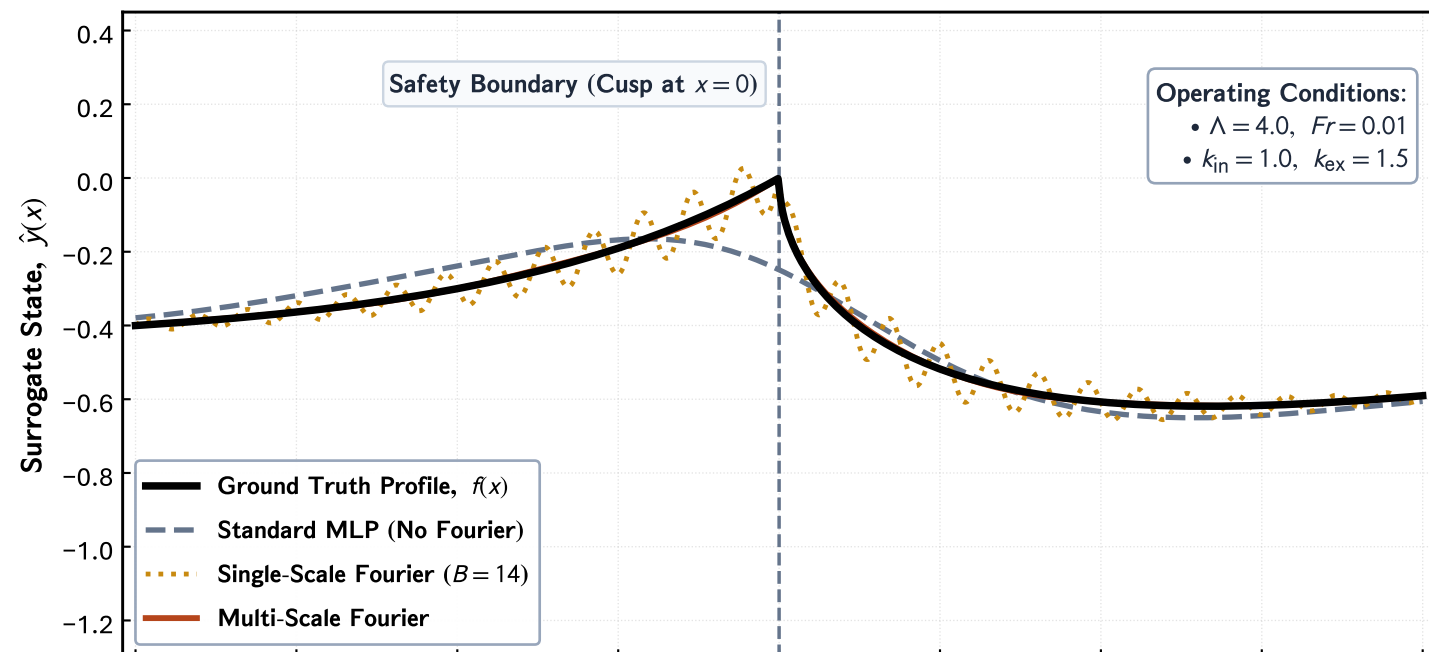
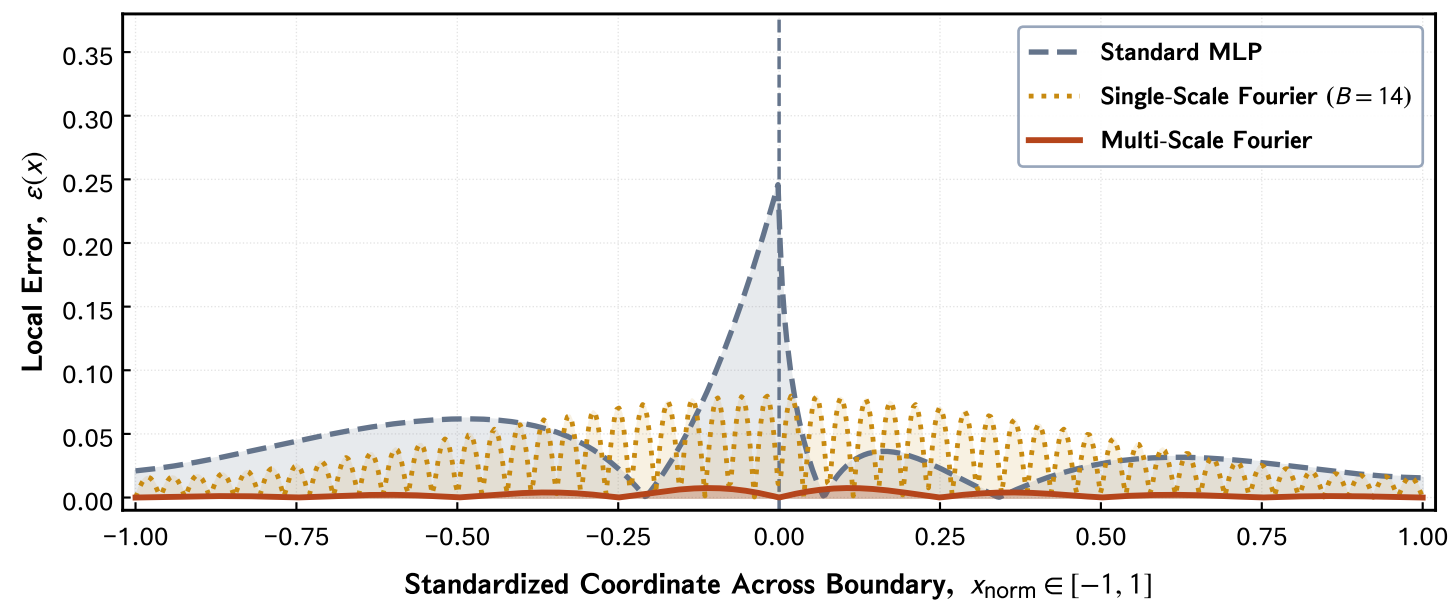
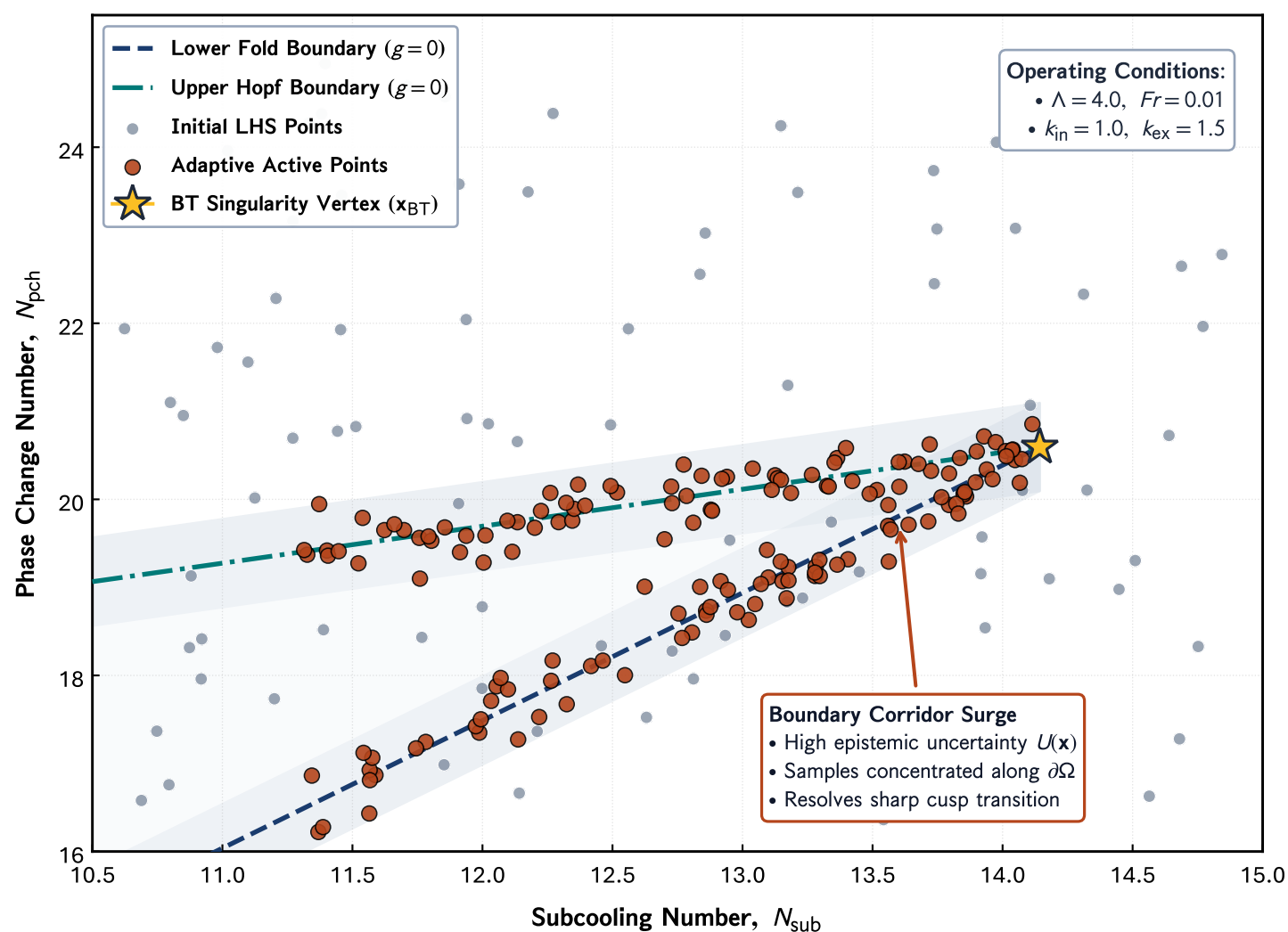
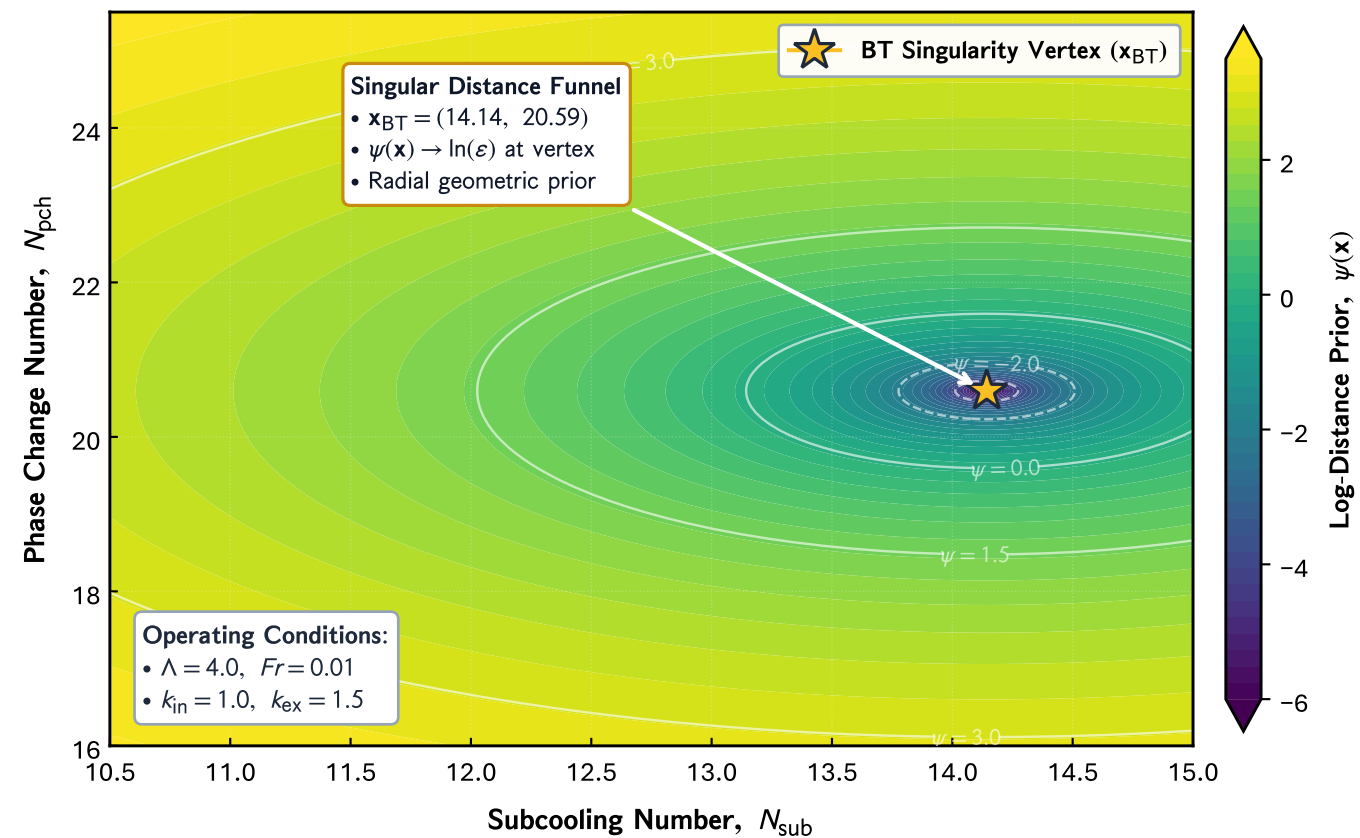
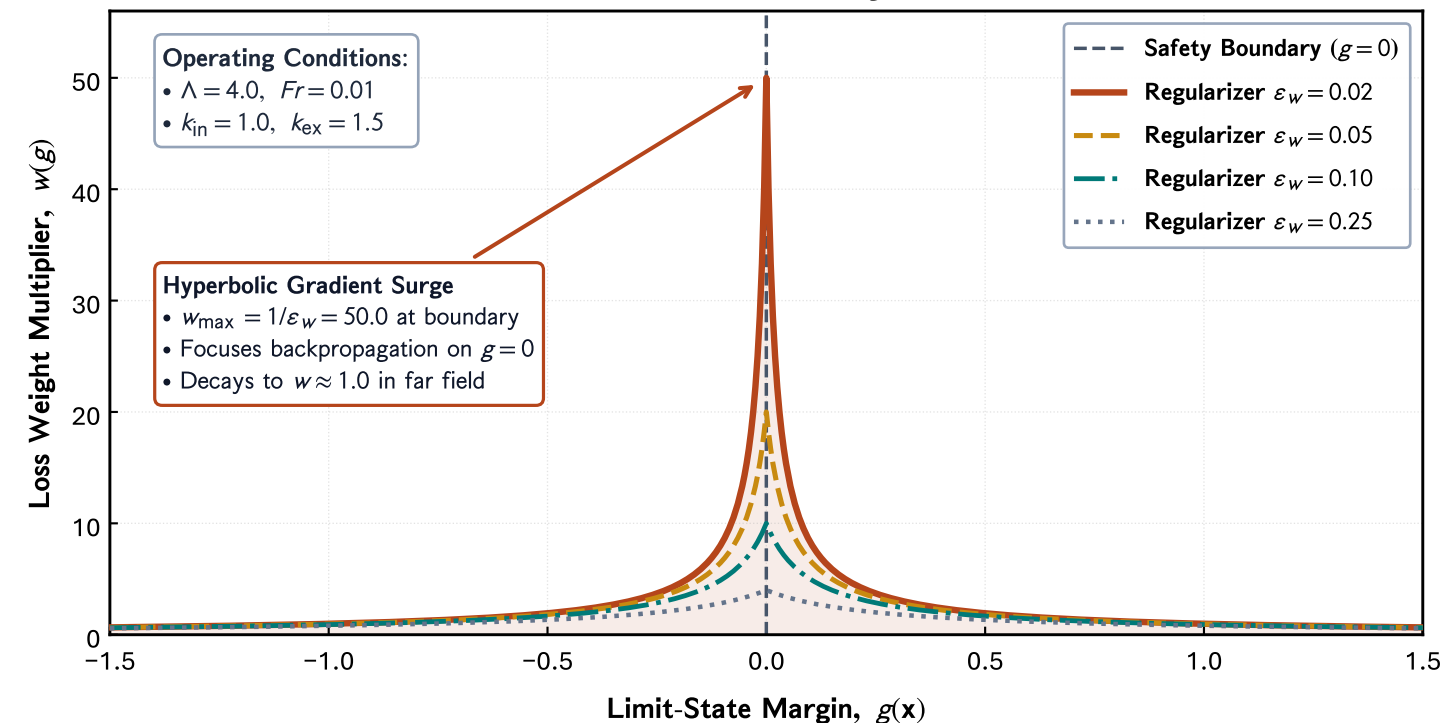


(a1) Boundary State Reconstruction : Spectral Bias vs. Multi-Scale Fourier

(a2) Local Prediction Error (Impact on Boundary Fidelity) :  $|\hat{y}(x) - \hat{f}(x)|$ (c) Misfit-Driven Adaptive Active Learning Allocation :  $\mathbf{x}_{k+1} = \arg \max [U(\mathbf{x}) \cdot \mathbf{I}(|g| \leq \delta_w)]$ (b) Analytical Log-Distance Coordinate Embedding :  $\psi(\mathbf{x}) = \ln(\|\mathbf{x} - \mathbf{x}_{\text{BT}}\|^2 + \varepsilon)$ (d) Inverse-Distance Boundary Loss Weighting :  $w(g) = \frac{1}{|g| + \varepsilon_w}$ (e) 2D Spatial Boundary Loss Weight Field :  $w(N_{\text{sub}}, N_{\text{pch}})$ 